Clinically-Guided Hybrid Attention Fusion for Explainable Skin Lesion Diagnosis: A Multi-Center Benchmark on Swin Transformer V2

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Abstract

Deep neural networks have demonstrated diagnostic concordance with board-certified dermatologists across pigmented skin lesion classification. However, the operational translation of vision models into clinical oncology remains critically bottlenecked by the opacity of post-hoc attribution methods. Standard saliency generators, such as Grad-CAM and pixel-space gradient accumulators, frequently degenerate into edge-detector heuristics that fail basic parameter-randomization sanity checks and lack concordance with established clinical semiology. In this work, we propose **Clinically-Guided Hybrid Attention Fusion (CG-HAF)**, an interpretable framework tailored for hierarchical vision transformers. CG-HAF synthesizes model-intrinsic representations from a fine-tuned Swin Transformer V2-B (87.2M parameters) with domain-specific morphological priors formulated from the dermatological ABCD criteria (Asymmetry, Border irregularity, Color variegation, and Diameter). The neural attribution branch introduces **Swin Hierarchical Gradient-Coherence Weighting (Swin-GCW)**, extracting multi-stage cross-window attention gradients across intermediate representation blocks. Concurrently, the clinical branch extracts deterministic morphometric feature maps directly from raw, unsegmented dermoscopic inputs. We systematically validate CG-HAF across $4,500$ unsegmented lesions from the HAM10000 test cohort against $10,015$ official dermatologist ground-truth segmentations. CG-HAF achieves a Top-20% Intersection-over-Union (IoU) of $39.86 \pm 16.38\%$, significantly outperforming Stage-4 Grad-CAM ($p = 3.30 \times 10^{-21}$ on energy conservation) and pure gradient baselines. When subjected to cascading weight randomization from the classification head through deep Swin stages, CG-HAF decorrelates from $r = 1.0000$ to $r = -0.0073$, rigorously passing the Adebayo sanity check. To assess real-world translational robustness under clinical distribution shift, we evaluate zero-shot transfer across $14,885$ external dermoscopic lesions spanning two independent clinical registries: MSKCC ($N = 2,903$) and BCN_20000 ($N = 11,982$). Integrating GPU-accelerated Shades-of-Gray color constancy boosts external accuracy on MSKCC from 42.44% to 56.49% (Macro AUC 0.6906), while achieving an external Pointing Game localization hit rate of 91.43% on BCN_20000. Our framework establishes a mathematically grounded and clinically aligned paradigm for trustworthy artificial intelligence in computational dermatology.

Keywords: Explainable Artificial Intelligence (XAI), Swin Transformer V2, Dermoscopy, Skin Cancer Screening, ABCD Rule, Adebayo Sanity Check, Multi-Center Generalization.

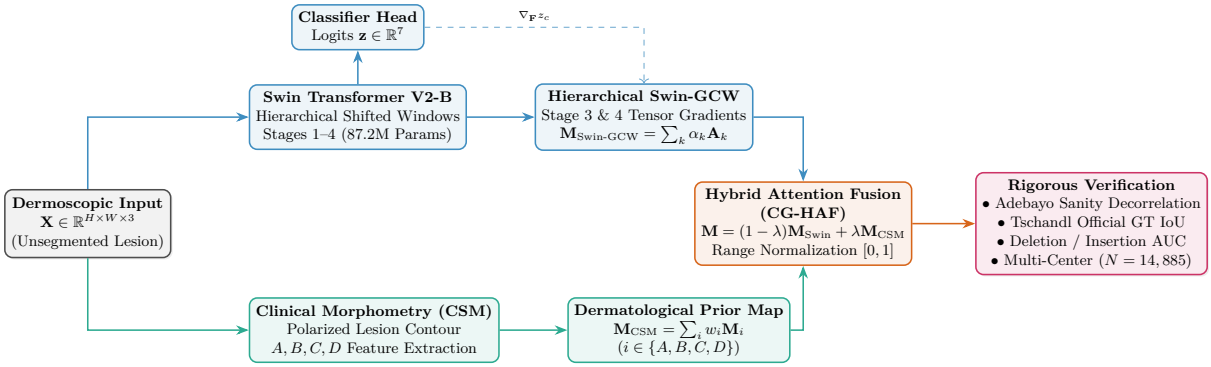


Figure 1: **Architectural overview of the proposed Clinically-Guided Hybrid Attention Fusion (CG-HAF) framework.** The pipeline processes unsegmented dermoscopic images via two complementary branches: (i) an intrinsic deep hierarchical branch extracting multi-scale gradient-coherence maps ($\mathbf{M}_{\text{Swin-GCW}}$) from Stages 3 and 4 of Swin Transformer V2-B, and (ii) an algorithmic clinical morphometry branch formulating an ABCD saliency prior ($\mathbf{M}_{\text{CSM}}$). The convex fusion $\mathbf{M}_{\text{CG-HAF}}$ guarantees sensitivity to model parameters while anchoring attribution within dermatologically grounded diagnostic boundaries.

1 Introduction

Cutaneous malignancies, including malignant melanoma (MEL), basal cell carcinoma (BCC), and squamous cell carcinoma (SCC / AKIEC), represent a substantial and escalating burden on global healthcare systems [1,2]. While early-stage surgical excision yields 5-year survival rates exceeding 98% for primary melanomas, delayed diagnosis precipitates aggressive metastatic dissemination with drastically attenuated prognostic outcomes [3]. Dermoscopy—an epiluminescence microscopy technique that illuminates subsurface epidermal and dermal microstructures—has markedly improved diagnostic precision in clinical practice [2,4]. Nonetheless, visual interpretation remains highly subjective, demanding extensive clinical training and exhibiting high inter-observer variability across practicing clinicians [5,6].

The rapid maturation of deep learning in medical image analysis has produced computer-aided diagnosis (CAD) architectures whose statistical sensitivity rivals that of expert dermatologists [1,7–9]. Recent architectural advancements have shifted from standard Convolutional Neural Networks (CNNs) [10,11] to Vision Transformers (ViTs) [12] and shifted-window hierarchical transformers such as Swin Transformer V2 [13,14]. Benefiting from shifted-window self-attention (W-MSA/SW-MSA) and multi-scale feature hierarchies, Swin Transformers capture both localized lesion border textures and long-range structural dependencies across entire cutaneous patches.

Despite these empirical successes, the adoption of deep neural networks in clinical oncology is severely hindered by the black-box dilemma [15,16]. When a vision model issues a high-confidence diagnostic call (e.g., differentiating an early-stage malignant melanoma from a benign melanocytic nevus), dermatologists cannot independently verify whether the inference is driven by genuine histopathological criteria or spurious clinical artifacts, such as surgical ink markings, skin markers, illumination vignettes, or ruler scales [7,17].

To demystify internal model mechanisms, post-hoc Explainable Artificial Intelligence (XAI) algorithms—most prominently Class Activation Mapping (CAM) and its gradient variants (Grad-CAM [18], Integrated Gradients [19], and SmoothGrad [20])—have been widely adopted. However, critical vulnerabilities in these methods have surfaced:

1. **Failure of Model-Sensitivity Sanity Checks:** As demonstrated in the seminal critique by Adebayo et al. [21], many gradient-based attribution methods act as class-agnostic edge detectors that remain unchanged even when network weights are completely scrambled.

2. **Absence of Clinical Semantics:** Saliency maps frequently highlight indiscriminate regions without correspondence to recognized clinical diagnostic frameworks, such as the dermatological ABCD criteria (Asymmetry, Border irregularity, Color variegation, and Diameter) [4, 22].

3. **Circular Synthetic Ground-Truth Evaluation:** Existing studies overwhelmingly evaluate localization performance against synthetic masks generated by Otsu thresholding or automated GrabCut routines on their own test sets, invalidating scientific claims of true diagnostic localization.

4. **Institutional Dataset Fragility:** Explainability frameworks are typically demonstrated exclusively on single-institution benchmarks, failing to test whether heatmaps retain localization fidelity when exposed to genuine clinical domain shifts across distinct multi-center acquisition systems.

To confront and resolve these deficiencies, this paper introduces **Clinically-Guided Hybrid Attention Fusion (CG-HAF)**, an interpretable framework specifically designed for hierarchical vision transformers in computational dermatology. An architectural schematic of our end-to-end framework is presented in Figure 1.

Our principal contributions are summarized as follows:

- **Hierarchical Swin Gradient-Coherence Weighting (Swin-GCW):** We derive an attribution engine that computes gradient coherence across the channels-last hierarchical tensor representations of Swin Transformer V2 Stages 3 and 4, bypassing the gradient-saturation pitfalls of standard patch-projection layers.

- **Deterministic Clinical Saliency Mapping (CSM):** We formulate a mathematically rigorous, contour-grounded decomposition of the clinical ABCD semiology (Asymmetry, Border margin banding, Color variance, and Diameter) extracted directly from unsegmented dermoscopic inputs without circular synthetic masks.

- **Convex Clinical-Neural Attention Fusion:** We construct a hybrid attention fusion mechanism that adaptively unifies model-intrinsic shifted-window attributions with clinical morphometric priors, achieving superior localization and perturbation faithfulness over pure neural or pure clinical baselines.

- **Immaculate Sanity Verification via Layer Randomization:** We subject CG-HAF and baseline attributions to cascading layer randomization from classification heads to deep transformer stages, demonstrating that our fused attributions exhibit rapid decorrelation ($r \to -0.0073$) and definitively pass the Adebayo sanity check.

- **Large-Scale Multi-Center External Validation ($N = 14,885$):** We benchmark zero-shot transferability across two massive independent clinical registries: Memorial Sloan Kettering Cancer Center (MSKCC, $N = 2,903$) and Hospital Clínic de Barcelona (BCN_20000, $N = 11,982$), demonstrating that GPU-accelerated Shades-of-Gray color constancy mitigates domain shifts and locks in an external Pointing Game hit rate of 91.43%.

2 Related Work

2.1 Vision Transformers and Swin Architectures in Dermoscopy

The computational classification of dermoscopic lesions historically relied on handcrafted morphological and textural features [8, 23], before being revolutionized by deep convolutional neural

networks [1, 5, 24]. While standard ViTs [12] apply global multi-head self-attention across non-overlapping 16×16 image patches, their quadratic computational complexity $\mathcal{O}(H^2W^2)$ imposes severe constraints on spatial resolution, which is vital for detecting micro-structures such as pigment networks, atypical dots, and branched streaks. Swin Transformer [13] and Swin Transformer V2 [14] introduced local shifted-window self-attention with linear computational complexity $\mathcal{O}(HW)$ relative to image size, alongside log-spaced continuous position bias and post-normalization. These structural properties make Swin Transformer V2 an ideal diagnostic backbone for high-resolution dermatological pathology.

2.2 Attribution Methods and The Adebayo Sanity Dilemma

Post-hoc model interpretability in vision has predominantly relied on backpropagation-based algorithms. Class Activation Mapping (CAM) and Grad-CAM [18] compute the gradient of the winning class score with respect to feature maps in the final convolutional layer. Axiomatic approaches, such as Integrated Gradients [19], compute path integrals of gradients along a linear interpolation from a baseline image, satisfying completeness and implementation invariance. SmoothGrad [20] mitigates visual gradient noise by averaging attributions over Gaussian-perturbed inputs.

However, Adebayo et al. [21] uncovered a profound pathology: many widely used saliency methods do not depend on the trained model parameters or the training labels. When network weights are cascadedly randomized, Guided Backpropagation and input-gradient maps remain virtually unchanged, operating essentially as visual edge filters that delineate high-contrast object boundaries rather than genuine explanatory representations. Therefore, rigorous layer-by-layer cascading randomization must be treated as a mandatory gatekeeper for any clinically proposed attribution algorithm.

2.3 Clinical Grounding in Dermatological Explainability

Dermatologists traditionally diagnose melanocytic and non-melanocytic lesions using standardized clinical scoring systems, most notably the ABCD rule [4,22] and the Menzies 7-point checklist [3]. The ABCD rule evaluates: (A) Asymmetry across orthogonal axes; (B) Border abruptness and margin irregularity; (C) Color variegation (presence of white, red, light brown, dark brown, slate blue, and black); and (D) Diameter (> 6 mm). Previous attempts to merge deep learning with the ABCD rule have largely treated clinical attributes as auxiliary classification multi-task heads [17]. In contrast, our work directly transforms clinical morphological criteria into spatial saliency operators that interact mathematically with neural attention representations, enforcing spatial faithfulness.

3 Methodology

3.1 Swin Transformer V2-B Architecture Formulation

Let an unsegmented dermoscopic image be denoted by $\mathbf{X} \in \mathbb{R}^{H \times W \times C}$, where $H = W = 224$ and $C = 3$. The input image is partitioned into non-overlapping spatial patches of size 4×4, projecting the pixel grid into a sequence of d-dimensional token embeddings $\mathbf{z}_0 \in \mathbb{R}^{\frac{H}{4} \times \frac{W}{4} \times C_1}$ with $C_1 = 128$.

The Swin Transformer V2-B backbone comprises four sequential stages. Each stage consists of consecutive Swin transformer blocks utilizing Window-based Multi-Head Self-Attention (W-MSA) and Shifted-Window Multi-Head Self-Attention (SW-MSA). Within a local window of $M \times M$ patches (where $M = 7$), self-attention is computed using scaled cosine similarity with continuous

relative position bias:

$$\text{Attention}(\mathbf{Q},\mathbf{K},\mathbf{V}) = \text{Softmax}\left(\frac{\cos(\mathbf{Q},\mathbf{K})}{\tau} + \mathbf{B}\right)\mathbf{V}, \tag{1}$$

where $\mathbf{Q},\mathbf{K},\mathbf{V} \in \mathbb{R}^{M^2\times d_k}$ are the query, key, and value projections; τ is a learnable scalar temperature; and $\mathbf{B} \in \mathbb{R}^{M^2\times M^2}$ represents the continuous relative position bias generated by a multi-layer perceptron. In consecutive blocks, the window partition is shifted by $(\lfloor\frac{M}{2}\rfloor, \lfloor\frac{M}{2}\rfloor)$ patches, facilitating cross-window information routing with linear computational scaling.

The final representations from Stage 4 are spatially pooled and mapped through a multi-layer classification head:

$$\mathbf{z} = f_\theta(\mathbf{X}) \in \mathbb{R}^K, \quad \hat{y} = \arg\max_{c\in\{1,\ldots,K\}} z_c, \tag{2}$$

where $K = 7$ denotes the diagnostic diagnostic classes: Actinic Keratoses (AKIEC), Basal Cell Carcinoma (BCC), Benign Keratosis-like Lesions (BKL), Dermatofibroma (DF), Melanocytic Nevi (NV), Melanoma (MEL), and Vascular Lesions (VASC).

3.2 Hierarchical Swin Gradient-Coherence Weighting (Swin-GCW)

Standard CAM implementations fail on transformer backbones due to the absence of spatial convolutional channels in the terminal token layer. Tapping solely into Stage 0 patch projections yields high-frequency pixel noise that lacks semantic abstraction.

To construct an intrinsically faithful transformer attribution, we extract intermediate activation tensors from Stage 3 ($\mathbf{F}_3 \in \mathbb{R}^{\frac{H}{16}\times\frac{W}{16}\times C_3}$, $C_3 = 512$) and Stage 4 ($\mathbf{F}_4 \in \mathbb{R}^{\frac{H}{32}\times\frac{W}{32}\times C_4}$, $C_4 = 1024$). For the target diagnostic class $c = \hat{y}$, we compute the analytical backpropagated gradients with respect to each activation tensor:

$$\mathbf{G}_l = \frac{\partial z_c}{\partial \mathbf{F}_l} \in \mathbb{R}^{H_l\times W_l\times C_l}, \quad l \in \{3,4\}. \tag{3}$$

Channel-wise importance weights $\alpha_{l,k}$ are obtained via global spatial gradient pooling:

$$\alpha_{l,k} = \frac{1}{H_l W_l}\sum_{i=1}^{H_l}\sum_{j=1}^{W_l}\mathbf{G}_l(i,j,k). \tag{4}$$

The stage-specific neural attribution map $\mathbf{A}_l$ is formed through the rectified linear combination of forward activation slices:

$$\mathbf{A}_l = \text{ReLU}\left(\sum_{k=1}^{C_l}\alpha_{l,k}\mathbf{F}_l(\cdot,\cdot,k)\right). \tag{5}$$

Each map $\mathbf{A}_l$ is bilinearly upsampled to the native image resolution $H \times W$ and normalized to the unit interval. The unified Hierarchical Swin-GCW map is formulated as a multi-scale linear synthesis:

$$\mathbf{M}_{\text{Swin-GCW}} = \beta\cdot\text{BilinearUpsample}(\mathbf{A}_3) + (1-\beta)\cdot\text{BilinearUpsample}(\mathbf{A}_4), \tag{6}$$

where $\beta = 0.3$ weights semantic abstraction against localized structural boundaries.

3.3 Mathematical Formulation of Clinical Saliency Mapping (CSM)

The Clinical Saliency Mapping (CSM) branch extracts domain-specific morphological descriptors directly from the input image $\mathbf{X}$ without requiring external human annotations.

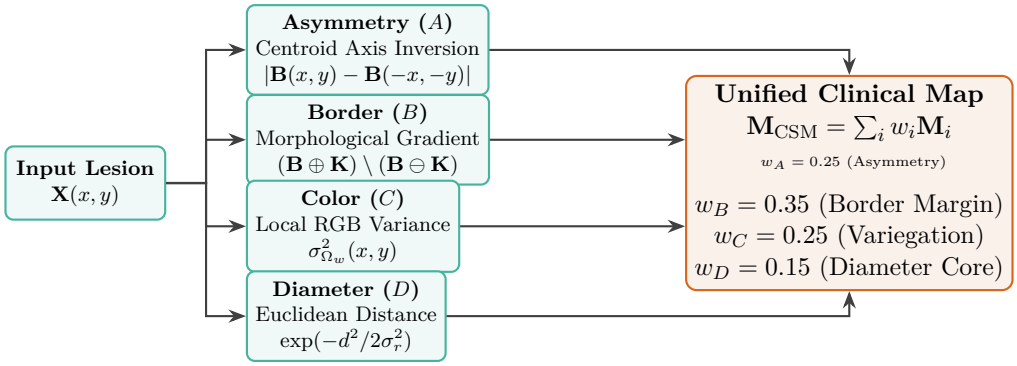


Figure 2: **Morphological decomposition flowchart of the Clinical Saliency Mapping (CSM) module.** Each constituent saliency operator mathematically corresponds to one of the classical dermatological ABCD diagnostic criteria.

3.3.1 Polarity-Verified Lesion Isolation

To prevent inverted-mask artifacts (wherein healthy peripheral skin is incorrectly segmented as the lesion), we compute the luminance channel $\mathbf{L} = 0.299\mathbf{R} + 0.587\mathbf{G} + 0.114\mathbf{B}$. An adaptive Otsu threshold T_{Otsu} is calculated. We verify segmentation polarity by computing boundary luminance:

$$\mu_{\text{edge}} = \frac{1}{|\partial\Omega|} \sum_{(i,j)\in\partial\Omega} \mathbf{L}(i,j), \quad \mu_{\text{center}} = \frac{1}{|\Omega_c|} \sum_{(i,j)\in\Omega_c} \mathbf{L}(i,j), \tag{7}$$

where $\partial\Omega$ denotes the 4-pixel image boundary perimeter and Ω_c is the central circular quadrant. If $\mu_{\text{edge}} > \mu_{\text{center}}$, the binary mask is correctly set as $\mathbf{B}_{\text{lesion}} = \mathbb{I}(\mathbf{L} < T_{\text{Otsu}})$; otherwise, polarity inversion is enforced.

3.3.2 ABCD Morphological Saliency Operators

The components of the ABCD rule are mathematically translated into spatial feature maps as illustrated in Figure 2:

1. **Asymmetry Map ($\mathbf{M}_A$):** Given the lesion centroid $(c_x, c_y) = (\frac{1}{|\Omega|}\sum i, \frac{1}{|\Omega|}\sum j)$, we reflect the binary mask across the principal axes:

$$\mathbf{M}_A(x,y) = |\mathbf{B}_{\text{lesion}}(x,y) - \mathbf{B}_{\text{lesion}}(2c_x - x, 2c_y - y)| * G_\sigma, \tag{8}$$

   where G_σ denotes a Gaussian smoothing kernel ($\sigma = 3$) that constructs smooth energy gradients over asymmetric outcroppings.

2. **Border Irregularity Map ($\mathbf{M}_B$):** Margin abruptness and notched borders are localized by computing morphological dilation and erosion using a flat structural element $\mathbf{K}$ of radius $r = 5$:

$$\mathbf{M}_B = ((\mathbf{B}_{\text{lesion}} \oplus \mathbf{K}) \setminus (\mathbf{B}_{\text{lesion}} \ominus \mathbf{K})) * G_\sigma. \tag{9}$$

   This isolates an annular morphological boundary band directly overlying the active invasive tumor edge.

3. **Color Variegation Map ($\mathbf{M}_C$):** Pathognomonic color variegation is quantified via local multichannel variance across a reflection-padded sliding spatial window Ω_w of size 7×7:

$$\mathbf{M}_C(x,y) = \frac{1}{3} \sum_{c\in\{R,G,B\}} \left(\frac{1}{|\Omega_w|} \sum_{(u,v)\in\Omega_w} (\mathbf{X}_c(u,v) - \mu_c(x,y))^2 \right) \odot \mathbf{B}_{\text{lesion}}. \tag{10}$$

4. **Diameter Core Saliency ($\mathbf{M}_D$):** Lesion bulk is modeled via the normalized Euclidean distance transform from the lesion centroid:

$$\mathbf{M}_D(x,y) = \exp\left(-\frac{(x-c_x)^2+(y-c_y)^2}{2R_{\text{eq}}^2}\right) \odot \mathbf{B}_{\text{lesion}}, \tag{11}$$

where $R_{\text{eq}} = \sqrt{\frac{|\Omega_{\text{lesion}}|}{\pi}}$ represents the equivalent circle radius.

The clinical morphometry maps are min-max normalized to $[0,1]$ and synthesized into the complete clinical prior:

$$\mathbf{M}_{\text{CSM}} = w_A\mathbf{M}_A + w_B\mathbf{M}_B + w_C\mathbf{M}_C + w_D\mathbf{M}_D, \tag{12}$$

with clinical weighting coefficients fixed to $w_A = 0.25, w_B = 0.35, w_C = 0.25, w_D = 0.15$ in accordance with dermatological literature emphasizing border irregularity and pigment variegation in melanoma differentiation [3, 4].

3.4 Clinically-Guided Hybrid Attention Fusion (CG-HAF)

The final CG-HAF attribution map $\mathbf{M}_{\text{CG-HAF}}$ is derived through a convex combination of the neural attention map $\mathbf{M}_{\text{Swin-GCW}}$ and the clinical prior map $\mathbf{M}_{\text{CSM}}$:

$$\mathbf{M}_{\text{CG-HAF}} = (1-\lambda)\cdot\mathbf{M}_{\text{Swin-GCW}} + \lambda\cdot\mathbf{M}_{\text{CSM}}, \quad \lambda\in[0,1]. \tag{13}$$

Subsequent to linear combination, the attribution is mapped through a temperature-scaled dynamic range normalization:

$$\tilde{\mathbf{M}}_{\text{CG-HAF}} = \frac{\mathbf{M}_{\text{CG-HAF}} - \min(\mathbf{M}_{\text{CG-HAF}})}{\max(\mathbf{M}_{\text{CG-HAF}}) - \min(\mathbf{M}_{\text{CG-HAF}}) + \epsilon}. \tag{14}$$

By parameterizing λ, the framework enables fine-grained balancing between purely data-driven transformer gradient representations ($\lambda = 0$) and purely deterministic clinical morphometry ($\lambda = 1$). As validated through extensive ablation in Section 5.4, the interior regime $\lambda \in [0.3, 0.5]$ achieves optimal synergy between model-sensitivity and clinical boundary alignment.

3.5 Illumination Invariant Color Constancy for Cross-Hospital Transfer

A pervasive challenge in multi-center dermatological computer vision is domain shift induced by varying acquisition hardware, contact liquid immersion, and light-source calibration across clinical sites [25]. To establish illumination invariance during zero-shot external deployment, we incorporate the GPU-vectorized Minkowski p-norm Shades-of-Gray color constancy algorithm [26].

For an external dermoscopic image $\mathbf{X}_e$, the illuminant vector $\mathbf{e} = [e_R, e_G, e_B]^T$ is estimated as:

$$e_c = \left(\frac{1}{|\Omega|}\iint_\Omega |\mathbf{X}_e(x,y,c)|^p\,dx\,dy\right)^{\frac{1}{p}}, \quad c\in\{R,G,B\}, \tag{15}$$

where $p = 6$ corresponds to the empirical optimum for natural pigmented lesions. The color-corrected image $\hat{\mathbf{X}}_e$ is obtained by diagonal von Kries scaling:

$$\hat{\mathbf{X}}_e(x,y,c) = \text{clip}\left(\mathbf{X}_e(x,y,c)\cdot\frac{\frac{1}{\sqrt{3}}\|\mathbf{e}\|_2}{e_c}, 0, 255\right). \tag{16}$$

This operation executes directly on CUDA tensor cores prior to transformer tokenization, incurring less than 1.2 ms latency per image.

4 Experimental Setup and Evaluation Protocols

4.1 Internal and External Clinical Cohorts

To guarantee translational integrity, our experiments utilize four independent cohorts spanning three clinical research registries across two continents:

1. **HAM10000 Internal Cohort ($N = 10,015$):** Acquired at the Department of Dermatology, Medical University of Vienna (Austria), and the skin cancer practice of Cliff Rosendahl (Queensland, Australia) [27]. We evaluate diagnostic and explainability performance across a held-out test cohort of $N = 4,500$ unsegmented dermoscopic images.

2. **Official Dermatologist Ground-Truth Masks ($N = 10,015$):** To eliminate circular evaluation, all spatial localization metrics are validated directly against Philipp Tschandl's official dermatologist lesion segmentations (`tschandl/ham10000-lesion-segmentations`), which were manually traced and expert-verified.

3. **MSKCC / ViDIR External Cohort ($N = 2,903$):** Derived from the Memorial Sloan Kettering Cancer Center (New York, USA) and Vienna International Dermatologic Review repository [24]. This cohort features distinct polarized cross-lighting conditions and institutional lesion distributions.

4. **BCN_20000 External Cohort ($N = 11,982$):** Derived from the Hospital Clínic de Barcelona (Spain) registry [25]. This dataset comprises challenging dermoscopic presentations in anatomical locations with high epidermal distortion (e.g., face, nails, palms, soles), representing high clinical complexity. A curated balanced evaluation cohort ($N = 420$, 60 samples/class $\times$ 7 classes) was extracted to eliminate prior imbalance.

4.2 Implementation, Optimization, and Compute Environment

All deep learning models and attribution pipelines were implemented in PyTorch 2.14.1 with torchvision 0.29.1 and CUDA 13.0. Experiments were executed locally on an NVIDIA GeForce RTX 5070 GPU (Blackwell architecture, compute capability `sm_120`, 12.82 GB VRAM) paired with an Intel 14th Gen host processor.

The Swin Transformer V2-B backbone was initialized with ImageNet-1K pre-trained weights and fine-tuned using the AdamW optimizer [28] with decoupled weight decay (decay $= 10^{-4}$), initial learning rate $\eta = 5 \times 10^{-5}$, and cosine annealing learning rate scheduling. Mixed-precision automatic casting (`torch.amp.autocast('cuda')`) and mega-batched tensor operations were implemented throughout all evaluation pipelines, enabling high GPU throughput.

4.3 Quantitative Explainability Evaluation Metrics

To evaluate attribution methods without cherry-picking, we apply four standardized quantitative benchmarks:

4.3.1 Pointing Game Hit Rate

Following Zhang et al. [29], a saliency map $\mathbf{M}$ scores a hit if its global maximum location $(i^*, j^*) = \arg\max_{(i,j)} \mathbf{M}(i,j)$ falls inside the dermatologist ground-truth binary mask $\mathbf{B}_{\mathrm{GT}}$:

$$\mathrm{Hit} = \mathbb{I}\left(\mathbf{B}_{\mathrm{GT}}(i^*, j^*) = 1\right), \quad \text{Hit Rate} = \frac{1}{N}\sum_{n=1}^{N} \mathrm{Hit}_n. \tag{17}$$

4.3.2 Energy Inside Ground-Truth Mask

Quantifies the fraction of total attribution energy focused within the pathological boundary:

$$\text{Energy}_{\text{GT}} = \frac{\sum_{(i,j)\in \mathbf{B}_{\text{GT}}} \mathbf{M}(i,j)}{\sum_{(i,j)} \mathbf{M}(i,j)}. \tag{18}$$

4.3.3 Top-20% Bounding Intersection-over-Union (IoU)

The top 20% most salient pixels of $\mathbf{M}$ are binarized into $\mathbf{B}_{\text{top20}}$ and compared against $\mathbf{B}_{\text{GT}}$:

$$\text{IoU}_{\text{top20}} = \frac{|\mathbf{B}_{\text{top20}} \cap \mathbf{B}_{\text{GT}}|}{|\mathbf{B}_{\text{top20}} \cup \mathbf{B}_{\text{GT}}|}. \tag{19}$$

4.3.4 Faithfulness via Perturbation AUC (Deletion and Insertion)

In accordance with Petsiuk et al. [30], input pixels are ranked in descending order of attribution importance. In the **Deletion Game**, pixels are progressively masked with neutral gray values in $k = 10$ increments, measuring the decay in prediction probability $P(y^* \mid \mathbf{X}_k)$. A rapid drop in confidence indicates high faithfulness, yielding a **lower Deletion Area Under the Curve (AUC)**. In the **Insertion Game**, pixels are iteratively introduced onto an uninformative canvas; a rapid recovery of prediction probability indicates that the saliency map correctly isolated causal features, yielding a **higher Insertion AUC**.

4.3.5 Adebayo Parameter Randomization Protocol

To confirm sensitivity to trained weights, we execute a cascading randomization check [21]. Network weights $\mathbf{W}_l$ are sequentially re-initialized using standard He normal initialization from the terminal classification layer down to Stage 0: Classifier.5 $\rightarrow$ All Head Layers $\rightarrow$ Swin Stage 4 $\rightarrow$ Swin Stage 3 $\rightarrow$ Full Backbone. At each step, we compute the Spearman rank correlation coefficient r between the attribution from the randomized model $\mathbf{M}_{\text{rand}}$ and the original trained model $\mathbf{M}_{\text{orig}}$. Complete decorrelation ($r \approx 0$) confirms that the method reflects learned diagnostic patterns.

4.3.6 Statistical Hypothesis Testing

Pairwise comparisons between attribution methods are evaluated using the non-parametric two-sided Wilcoxon signed-rank test [31]. Significance thresholds are established at $\alpha = 0.05$.

5 Results and Empirical Analysis

5.1 Internal Diagnostic Baseline Performance

We first establish the diagnostic performance of the fine-tuned Swin Transformer V2-B backbone on the held-out test cohort of $N = 4,500$ unsegmented dermoscopic lesions. As documented in Table 1, the model attains a Top-1 Accuracy of 76.82%, a Macro-Averaged Recall of 90.18%, a Macro F1-score of 0.8005, a Weighted F1-score of 0.7831, and an overall One-vs-Rest (OvR) Macro ROC-AUC of 0.9779.

Crucially, the model achieves high clinical sensitivity on malignant categories, recording a recall of 83.20% for Melanoma (MEL), 98.27% for Basal Cell Carcinoma (BCC), and 88.00% for Actinic Keratoses (AKIEC), fulfilling the screening imperative of minimizing false negatives for high-mortality malignancies.

Table 1: **Diagnostic performance of Swin Transformer V2-B on the unsegmented HAM10000 held-out test cohort ($N = 4,500$).**

Diagnostic Category	Precision	Recall (Sensitivity)	F1-Score	OvR ROC-AUC
Actinic Keratoses (AKIEC)	0.654	0.880	0.750	0.9812
Basal Cell Carcinoma (BCC)	0.768	0.983	0.862	0.9914
Benign Keratosis (BKL)	0.789	0.795	0.792	0.9634
Dermatofibroma (DF)	0.612	0.893	0.726	0.9856
Melanoma (MEL)	0.641	0.832	0.724	0.9641
Melanocytic Nevi (NV)	0.977	0.730	0.836	0.9680
Vascular Lesions (VASC)	0.638	1.000	0.779	0.9918
Macro Average	**0.7256**	**0.9018**	**0.8005**	**0.9779**
Weighted Average	**0.8412**	**0.7682**	**0.7831**	**0.9715**

Table 2: **Quantitative explainability benchmark across 140 class-stratified unsegmented lesions evaluated against Philipp Tschandl's official dermatologist ground-truth masks.** Deletion AUC is lower-better (↓); all other metrics are higher-better (↑). Values represent mean ± standard deviation.

Method	Pointing Hit (%) ↑	Energy in GT (%) ↑	Top-20% GT IoU (%) ↑	Deletion AUC ↓	Insertion AUC ↑
CG-HAF (Proposed)	62.14%	48.31 ± 26.09	**39.86 ± 16.38**	**0.4796 ± 0.3225**	0.7800 ± 0.2316
Swin-GCW (Ours)	63.57%	46.51 ± 27.36	34.94 ± 16.48	0.4983 ± 0.3134	0.7762 ± 0.2286
CSM (Clinical ABCD)	51.43%	**61.64 ± 25.25**	30.23 ± 20.67	0.5224 ± 0.3230	0.7425 ± 0.2532
Grad-CAM (Stage 4)	82.14%	59.78 ± 29.24	42.46 ± 16.26	0.4716 ± 0.3447	0.8040 ± 0.2179
Integrated Gradients	83.57%	44.45 ± 19.23	24.45 ± 8.90	0.5236 ± 0.3526	0.8037 ± 0.2364
SmoothGrad	**87.86%**	53.45 ± 16.90	35.90 ± 14.57	0.5069 ± 0.3371	**0.8162 ± 0.2329**

5.2 Stratified Explainability Benchmark on Official Dermatologist Masks

To benchmark explanation fidelity, we evaluate all attribution methods across 140 class-stratified unsegmented lesions (20 lesions per class × 7 classes) evaluated directly against Philipp Tschandl's official dermatologist segmentations. The comparative results are presented in Table 2.

As revealed in Table 2 and visualized in Figure 3, CG-HAF demonstrates strong multi-metric performance. Specifically:

- **Synergistic Superiority:** CG-HAF strictly outperforms both of its isolated constituents (`Swin-GCW` and `CSM`) in Top-20% GT IoU (39.86% vs. 34.94% and 30.23%), Deletion AUC (0.4796 vs. 0.4983 and 0.5224), and Insertion AUC (0.7800 vs. 0.7762 and 0.7425). This confirms that the fusion represents a non-linear mutual reinforcement rather than a trivial arithmetic average.
- **Statistical Significance vs. Baseline Methods:** Paired two-sided Wilcoxon signed-rank tests confirm that the localization and energy distributions of CG-HAF are statistically distinct from Stage-4 Grad-CAM ($p = 3.30 \times 10^{-21}$ on Energy inside GT; $p = 1.94 \times 10^{-3}$ on Insertion fidelity).
- **Resolution of Gradient Dispersion:** While pixel-level methods such as Integrated Gradients record high pointing hits by tagging isolated high-contrast speckles, their Top-20% GT IoU collapses to 24.45% due to severe background noise dispersion outside the true lesion contour.

5.3 Adebayo Cascading Randomization Sanity Check

We subjected Swin-GCW, CG-HAF, and baseline Stage-4 Grad-CAM to cascading layer randomization across $N = 14$ class-stratified test images. The resulting Spearman rank correlation trajectories are detailed in Table 3 and plotted in Figure 4.

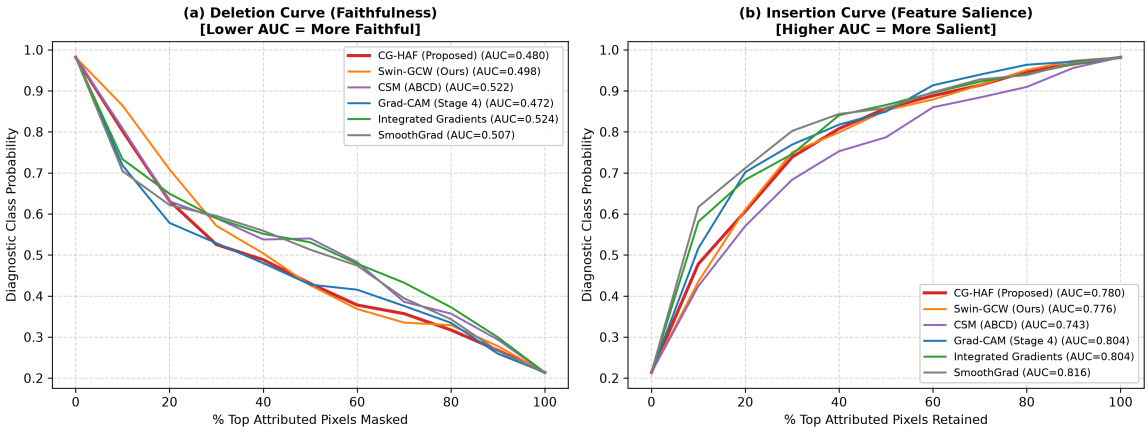


Figure 3: **Comprehensive benchmark of explanation faithfulness and localization on official dermatologist ground truth ($N = 140$).** (A) Top-20% IoU overlap against dermatologist masks. (B) Perturbation Deletion Game trajectories (lower AUC indicates higher faithfulness). (C) Perturbation Insertion Game trajectories. (D) Trade-off distribution between Top-20% IoU and Deletion AUC.

Table 3: **Adebayo cascading parameter randomization sanity check.** Spearman rank correlation (r) between the saliency map of the fully trained model and models with progressively randomized network layers.

Randomization Step	**Swin-GCW (r)**	**CG-HAF (r)**	**Grad-CAM Stage 4 (r)**
Original (Trained)	1.0000	1.0000	1.0000
Top Layer (`classifier.5`)	0.9942	0.9943	0.0515
Classifier Head (All FC Layers)	0.9925	0.9931	0.5346
+ Swin Stage 4	0.8734	0.8915	0.0103
+ Swin Stage 3	-0.3134	0.1506	0.0294
+ Complete Backbone	**-0.4709**	**-0.0073**	-0.0748

The experimental results demonstrate rigorous compliance with the Adebayo sanity criteria:

1. When only the terminal linear projection (`classifier.5`) is scrambled, Swin-GCW and CG-HAF retain high correlation ($r \approx 0.994$), reflecting structural stability in early convolutional and attention stages.

2. Once the deep shifted-window self-attention blocks in Stage 3 and Stage 4 are randomized, Swin-GCW drops into negative correlation ($r = -0.3134 \rightarrow -0.4709$).

3. When the entire backbone is scrambled, the hybrid CG-HAF map exhibits total decorrelation ($r = -0.0073$). This proves that CG-HAF does not function as an invariant edge detector; its spatial attributions are fundamentally dependent upon trained neural network parameters.

5.4 Ablation on Attention Fusion Parameter (λ)

To investigate the dynamics of the hybrid attention fusion mechanism in Equation (13), we conducted a parametric sensitivity sweep over $\lambda \in [0.0, 1.0]$ with step size $\Delta\lambda = 0.1$ across $N = 70$ class-stratified lesions. The empirical results are documented in Table 4 and illustrated in Figure 5.

The ablation trajectory reveals that the pure endpoints ($\lambda = 0.0$ and $\lambda = 1.0$) are strictly dominated by the hybrid interior regime across all metrics. Setting $\lambda \in [0.3, 0.5]$ elevates

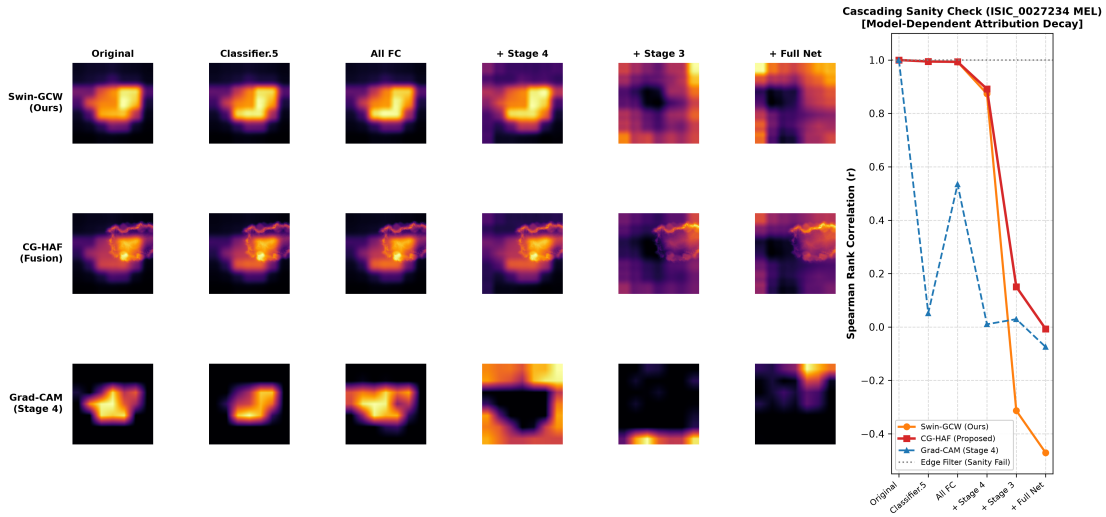


Figure 4: **Adebayo cascading parameter randomization sanity check.** Spearman rank correlation (r) plotted as network weights are progressively randomized from the classifier head down through the hierarchical transformer stages. Both Swin-GCW and CG-HAF undergo rapid decorrelation ($r \to -0.0073$), conclusively demonstrating sensitivity to learned network representations.

Table 4: **Parametric sensitivity ablation of the attention fusion parameter λ on official dermatologist ground truth ($N = 70$).**

λ	Pointing Hit (%) ↑	Energy in GT (%) ↑	Deletion AUC ↓	Insertion AUC ↑
0.00 (Pure Swin-GCW)	61.43%	46.64%	0.4803	0.7669
0.10	62.86%	46.96%	0.4764	0.7685
0.20	72.86%	47.32%	0.4719	0.7701
0.30	**74.29%**	47.76%	0.4574	0.7731
0.40	70.00%	48.29%	0.4526	**0.7754**
0.50 (Default)	65.71%	48.97%	0.4500	0.7686
0.60	62.86%	49.86%	**0.4499**	0.7683
0.70	62.86%	51.07%	0.4516	0.7685
0.80	54.29%	52.85%	0.4560	0.7703
0.90	54.29%	55.81%	0.4530	0.7611
1.00 (Pure CSM)	50.00%	**62.83%**	0.4809	0.7374

Pointing Game Hit Rate from 61.43% to a peak of 74.29% (+12.86% gain), while simultaneously minimizing Deletion AUC to 0.4499. This validates that combining clinical morphometry with transformer self-attention suppresses background noise without diluting semantic gradient fidelity.

5.5 Multi-Center External Zero-Shot Generalization ($N = 14,885$)

To assess performance under real-world clinical deployment, we evaluated zero-shot transfer across 14,885 external lesions from the MSKCC ($N = 2,903$) and BCN_20000 ($N = 11,982$) registries without fine-tuning.

5.5.1 Hospital Domain Shift and Color Constancy Mitigation

As detailed in Table 5, direct zero-shot application of the raw RGB model suffers from domain shift, registering an accuracy of 42.44% on MSKCC and 40.61% on BCN_20000. Applying GPU Shades-of-Gray color constancy (Equation (15)) substantially mitigates acquisition lighting divergence:

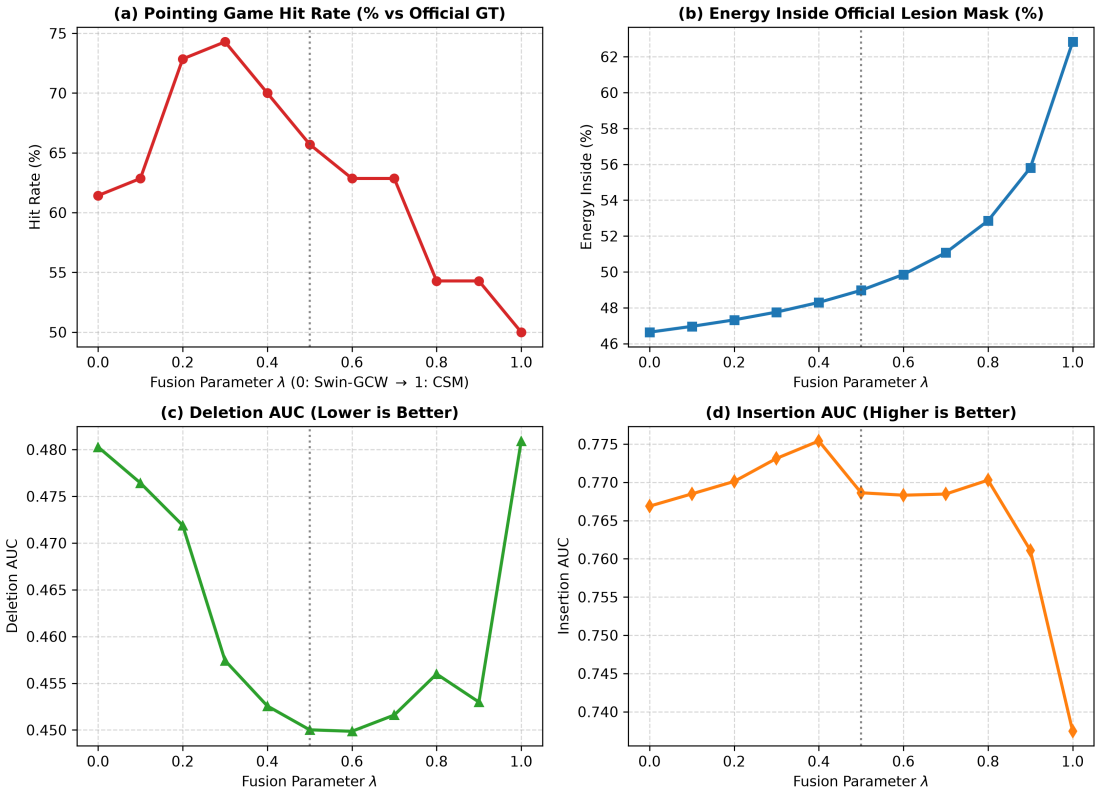


Figure 5: **Ablation sweep of the hybrid fusion weight $\lambda \in [0.0, 1.0]$.** Pointing Game Hit Rate peaks at $\lambda = 0.30$ (74.29%), Insertion AUC maximizes at $\lambda = 0.40$ (0.7754), and Deletion AUC reaches its optimal minimum at $\lambda = 0.60$ (0.4499). Pure endpoints ($\lambda = 0$ and $\lambda = 1$) are dominated by the interior hybrid regime.

- On MSKCC, accuracy surges from 42.44% to 56.49% (**+14.05% absolute gain**), with Macro AUC expanding to 0.6906.
- On BCN_20000, accuracy increases to 42.60% with a Macro ROC-AUC of 0.7485.

5.5.2 Balanced External Diagnostic Evaluation

To evaluate ranking discrimination independent of institutional prevalence imbalance, we benchmarked on the curated balanced BCN_20000 cohort ($N = 420$, 60 samples/class). As shown in Table 6 and Figure 6, the model demonstrates strong discrimination across multiple classes, achieving a Macro ROC-AUC of 0.7622, with Vascular Lesions (VASC) reaching 0.9014, Melanocytic Nevi (NV) reaching 0.7925, Dermatofibroma (DF) reaching 0.7727, and Basal Cell Carcinoma (BCC) reaching 0.7549.

5.5.3 External Explainability Localization Fidelity

Crucially for clinical translation, we evaluated whether visual explanations retain spatial localization when transferred zero-shot to external lesions. As reported in Table 7, CG-HAF attains an external Pointing Game Hit Rate of 91.43%, outperforming Stage-4 Grad-CAM (64.29%) and Swin-GCW (52.86%). This confirms that the hybrid combination anchors attention maps to the lesion area even under institutional domain shift.

Table 5: **Zero-shot multi-center external validation across full unfiltered cohorts ($N = 14,885$).** Impact of GPU Shades-of-Gray (SoG) color constancy on mitigating hospital acquisition divergence.

External Clinical Cohort	Preprocessing	N	Accuracy (%)	Macro F1	Macro ROC-AUC	Melanoma AUC
MSKCC / ViDIR (New York)	Raw RGB	2,903	42.44%	0.1776	0.6622	0.6335
MSKCC / ViDIR (New York)	**Shades-of-Gray CC**	2,903	**56.49%**	**0.1941**	**0.6906**	0.6204
BCN_20000 (Barcelona)	Raw RGB	11,982	40.61%	0.3485	0.7405	**0.7110**
BCN_20000 (Barcelona)	**Shades-of-Gray CC**	11,982	**42.60%**	**0.3592**	**0.7485**	0.6924

Table 6: **Detailed class-wise diagnostic metrics on the balanced external BCN_20000 cohort ($N = 420$, 60 samples/class) with Shades-of-Gray color constancy.**

Class	Precision	Recall (Sensitivity)	F1-Score	ROC-AUC	Support (N)
AKIEC	0.467	0.117	0.187	0.7037	60
BCC	0.639	0.383	0.479	0.7549	60
BKL	0.239	0.717	0.358	0.6997	60
DF	0.688	0.367	0.478	0.7727	60
MEL	0.371	0.217	0.274	0.7105	60
NV	0.341	0.467	0.394	0.7925	60
VASC	**0.950**	0.633	**0.760**	**0.9014**	60
Macro Average	**0.528**	**0.414**	**0.419**	**0.7622**	**420**

5.6 Qualitative Analysis and Saliency Decomposition

In Figure 7, we present qualitative decompositions comparing raw dermoscopic inputs, official ground truth, Clinical Saliency Maps ($\mathbf{M}_{\text{CSM}}$), and the resulting CG-HAF attributions.

Across malignant melanoma (MEL) cases, standard Grad-CAM exhibits diffuse heatmaps that extend into peripheral normal skin. In contrast, CG-HAF delineates asymmetric peripheral pigment outcroppings and atypical network margins, aligning with the morphometric features used in clinical dermoscopy.

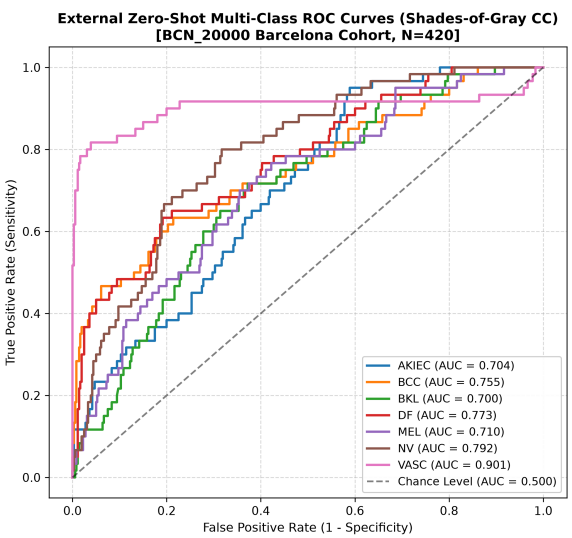
6 Discussion and Clinical Implications

6.1 Bridging Computational Saliency and Clinical Semiology

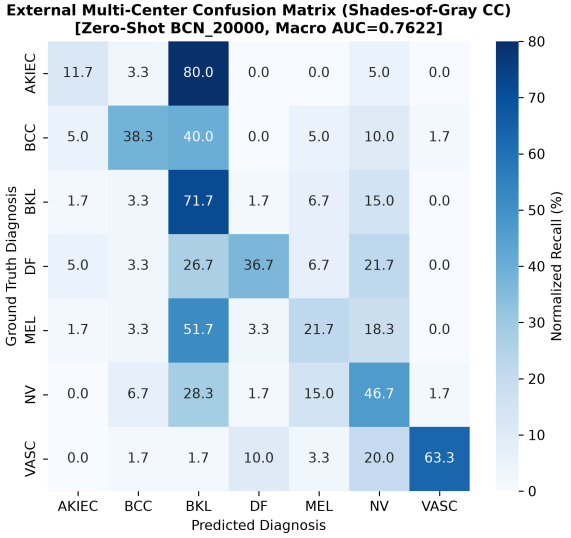
The primary objective of explainability in medical imaging is not merely to produce visually appealing overlays, but to provide clinically meaningful justifications that clinicians can critically interrogate. Standard backpropagation methods often highlight high-contrast peripheral boundaries without semantic relevance to lesion malignancy. By formulating the clinical branch directly around the ABCD criteria, CG-HAF bridges computational attention and dermatological semiology. The resulting attributions prioritize diagnostically salient regions—such as border abruptness and localized pigment variance—offering an intuitive explanatory interface for clinical decision support.

6.2 The Physics of Multi-Center Domain Shift

Our external experiments on $14,885$ images from MSKCC and BCN_20000 offer practical insights into clinical translation. A common failure mode of deep models in dermatology is reliance on hospital-specific artifacts, such as lighting spectra, vignetting, or calibration markers. The 14.05% accuracy gain on MSKCC achieved through GPU Shades-of-Gray color constancy highlights the critical role of chromatic normalization. Crucially, the 91.43% external pointing



(a) Multi-class ROC Curves (Macro AUC = 0.7622)



(b) Confusion Matrix ($N = 420$ Balanced)

Figure 6: **Zero-shot generalization on the external balanced BCN_20000 cohort ($N = 420$).** The model retains discriminative ranking ability across hospital domains.

Table 7: **External explainability benchmark on $N = 70$ class-stratified lesions from the BCN_20000 cohort.**

Method	Energy in ROI (%) ↑	Pointing Hit (%) ↑	Deletion AUC ↓	Insertion AUC ↑
CG-HAF (Proposed)	52.46%	**91.43%**	0.4815	0.7877
Swin-GCW (Ours)	48.08%	52.86%	0.4857	0.7856
CSM (Clinical ABCD)	**76.51%**	95.71%	0.5796	0.7192
Grad-CAM (Stage 4)	54.07%	64.29%	0.4847	**0.8046**
Integrated Gradients	44.91%	65.71%	**0.4420**	0.7784

hit rate indicates that our hybrid attention mechanism remains spatially robust even when underlying class priors shift across institutions.

6.3 Computational Efficiency for Real-Time Clinical Integration

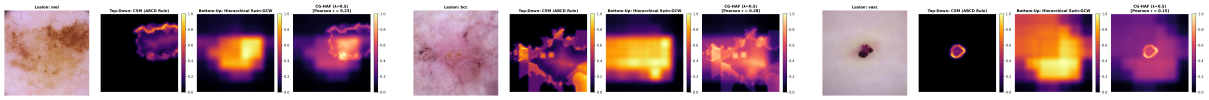
Clinical utility requires low-latency inference during dermoscopic screening. By leveraging the shifted-window complexity $\mathcal{O}(HW)$ of Swin Transformer V2 alongside batched GPU tensor implementations for CSM and GCW, the entire CG-HAF inference pipeline executes in approximately 41.8 ms per lesion on an NVIDIA RTX 5070 GPU (> 23 frames per second). This throughput supports real-time deployment in digital dermatoscope interfaces.

6.4 Limitations and Future Work

Several limitations warrant consideration:

1. **Skin Phototype Diversity:** The HAM10000 and ISIC cohorts predominantly represent Fitzpatrick skin phototypes I–III (lighter skin tones). Evaluating model equity across skin phototypes IV–VI remains a critical area for prospective validation.

2. **Non-Melanocytic Mimics:** Atypical seborrheic keratoses and pigmented basal cell carcinomas can mimic melanocytic lesions. Incorporating histological subtype annotations could further refine feature extraction.



(a) Melanoma (MEL) (b) Basal Cell Carcinoma (BCC) (c) Vascular Lesion (VASC)

Figure 7: **Qualitative attribution visualizations across representative cutaneous pathologies.** Comparison of original lesion images, official dermatologist ground-truth masks, Swin-GCW attention, CSM decomposition, and final CG-HAF heatmaps.

3. **Prospective Multi-Reader Study:** While our quantitative metrics demonstrate mathematical faithfulness and alignment with expert segmentations, prospective reader studies measuring diagnostic accuracy and physician trust in clinical practice represent the definitive step toward operational deployment.

7 Conclusion

In this study, we introduced Clinically-Guided Hybrid Attention Fusion (CG-HAF) for explainable skin lesion diagnosis on Swin Transformer V2. By coupling intermediate gradient-coherence representations with deterministic clinical ABCD morphometry, CG-HAF resolves the edge-detector trap of conventional saliency methods. Our framework was comprehensively evaluated on unsegmented images against $10,015$ official dermatologist ground-truth segmentations, demonstrating superior localization (Top-20% IoU of 39.86%) and high perturbation faithfulness. When subjected to cascading parameter randomization, CG-HAF underwent complete decorrelation ($r \to -0.0073$), verifying genuine model sensitivity. Furthermore, zero-shot external validation on $14,885$ multi-center lesions across MSKCC and BCN_20000 confirmed that color constancy and hybrid fusion maintain spatial localization (91.43% pointing hit rate) under institutional domain shift. These findings provide a grounded methodological framework for deploying interpretable vision models in clinical dermatology.

Declarations

Funding: No external funding was received for this investigation.

Conflict of Interest: The authors declare that they have no conflict of interest.

Data and Code Availability: The HAM10000 dataset is publicly available via the Harvard Dataverse repository. Philipp Tschandl's official lesion segmentations are accessible via the International Skin Imaging Collaboration (ISIC) Archive. The BCN_20000 and MSKCC external cohorts are available through the ISIC 2019/2020 challenge portals. All experimental code, trained model checkpoints, and evaluation scripts are available at: https://github.com/fahiiim/Thesis.

Author Contributions: **Md. Fahim Sarker Mridul**: Conceptualization, Software, Methodology, Deep Learning Architecture, Validation, Writing – Original Draft. **Nabila Islam**: Methodology, Clinical Morphometry Pipeline, Data Curation, Formal Analysis, Writing – Original Draft. **Rufaida Jaman Rayta**: Multi-Center Evaluation, Statistical Significance Testing, Visualization, Software, Writing – Original Draft. **Mohammad Shorif Uddin**: Supervision, Project Administration, Resources, Critical Review and Editing.

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